

A century and a half of bear markets

Depth, duration, and recovery across fifteen U.S. equity drawdowns, 1871–2022

ABSTRACT

We examine the universe of S&P 500 bear markets over 152 years using inflation-adjusted total returns and a 20% drawdown threshold. We identify fifteen distinct episodes between 1871 and 2022. Median peak-to-trough depth is 31.7%, median duration is 18 months, and median recovery to prior real high is 24 months. The distribution is, however, severely right-skewed in all three dimensions: the 1929–32 collapse reached 76.8% in real terms, the 1973–74 episode took 121 months to recover, and the 2000–09 episode lasted 103 months from peak to trough. We document that recovery time correlates more strongly with the inflation regime at the trough (Spearman $\rho = 0.62$ with average trough-period CPI) than with drawdown depth itself ($\rho = 0.41$). Bear markets occurring during high inflation regimes recover materially more slowly than equally deep bears in stable-price regimes, a finding with direct implications for asset allocation through cycles. We close by examining the still-unfolding 2022 bear market against this historical reference.

KEY FINDINGS

- 01 Fifteen real bears in 152 years.** Using a 20% real-total-return drawdown threshold, the S&P 500 experienced fifteen distinct bear markets between 1871 and 2022, or roughly one per decade.
- 02 Median bear is shallower than headline statistics suggest.** Median peak-to-trough decline is 31.7% in real terms, with 18-month duration and 24-month recovery — well inside what disciplined investors can absorb.
- 03 The right tail is what kills you.** The 1929–32 episode reached 76.8%, the 2000–09 episode took 103 months peak-to-trough, and the 1973–74 episode took ten years to recover.

- 04 **Inflation determines recovery speed.** Recovery time correlates 0.62 with the inflation regime at the trough — a stronger relationship than with drawdown depth itself (0.41).
- 05 **2022 sits below the median.** The 2022 drawdown reached 24.5% in real terms over 11 months — milder and shorter than the historical median, but in the most hostile inflation regime since 1980.

EXEC Executive summary

Bear markets are conventionally taught as exceptions to the long-run upward drift of equity returns. Examined across 152 years, they look more like a recurring feature of the system: regular enough to plan around, irregular enough to defeat simple rules.

This note documents the full bear-market history of the U.S. equity market from 1871 to October 2022 using Robert Shiller's inflation-adjusted total return series. We define a bear market as any peak-to-trough decline of at least 20% in real total return terms, with the episode considered closed when the index recovers to its prior real high. The threshold and recovery definition are deliberately conservative: nominal-return definitions count fewer episodes, but obscure the wealth-erosion impact of inflation during the worst bears.

We identify fifteen episodes meeting this definition. The full list spans 1876–77, 1892–93, 1902–03, 1906–07, 1911–14, 1916–20, 1929–32, 1937–42, 1946–48, 1961–62, 1968–70, 1973–74, 1987 (briefly real-terms over 20% inclusive of inflation), 2000–09, and the as-of-this-writing 2021–2022 episode.

The central finding is that the headline statistics of the average bear are reassuring, but those of the worst bears are devastating. Median depth is 31.7%, median duration 18 months, median recovery 24 months. These are numbers an investor with a long horizon and intact financial position can endure. But the worst quartile of episodes reached depths of 47%–77%, durations of 33–103 months, and recoveries of 36–121 months. An investor with the misfortune to begin a bear market at the wrong age, or with insufficient liquidity to hold through it, can face permanent damage to their financial trajectory.

The second finding is that what determines whether a bear market is a temporary setback or a generational event is, primarily, the inflation regime at the trough. Bears that resolved into deflation or low inflation (1932, 1942, 1962, 2009) recovered to new real highs within 24–53 months. Bears that resolved into elevated inflation (1920, 1948, 1974) took 32–121 months to recover. The 1973–74 bear, despite being only the seventh-deepest in our sample, holds the longest recovery time at ten years, because the 1970s inflation environment continued to erode the real value of even nominal price recoveries.

The third finding bears on the present moment. The 2022 drawdown, as of this writing in October 2022, has reached 24.5% in real terms over 11 months — shorter and shallower than the historical median. But it is occurring in the most hostile inflation regime since 1980, with year-over-year CPI peaking at 9.1%. If our cross-sectional finding holds, the recovery time for this episode will be determined less by current valuation levels (which have de-rated significantly) and more by whether inflation normalises or persists.

The note proceeds as follows. Section 2 defines our bear-market identification methodology and discusses alternatives. Section 3 reviews the relevant literature. Section 4 presents the full catalogue of episodes with summary statistics. Section 5 examines the inflation-recovery relationship. Section 6 places the 2022 episode in historical context. Section 7 discusses limitations and concludes.

01 Introduction

The phrase "bear market" obscures more than it reveals. Used in industry parlance to mean a 20% decline from peak in nominal price terms, the threshold collapses together episodes as different as the four-month 1987 crash, which an investor unburdened by leverage could have ignored, and the 2000–09 secular bear, which permanently damaged an entire generation's perception of equity ownership. The two events share a label and almost nothing else.

This is a problem for thinking. The mental category "bear market" suggests a unitary phenomenon, with characteristic features one can study, predict, and prepare for. The empirical reality is a heavy-tailed distribution of episodes that vary across at least three dimensions — depth, duration, and recovery — by an order of magnitude in each. The median bear and the worst bear are not points on a continuum so much as different objects.

This note addresses the problem by examining the full population of bear markets in U.S. equity history, classifying them along their three principal dimensions, and identifying which environmental factors predict where on the distribution a given episode lands. Our central conclusion — that inflation regime at the trough drives recovery time more reliably than drawdown depth — has practical implications: it suggests that two structurally similar drawdowns (say, the 1973 and 2000 cases, both initiated by valuation peaks) can produce dramatically different investor experiences depending on the macroeconomic regime in which they resolve.

The note has three secondary aims. First, to update the historical record through the as-of-this-writing 2022 episode. Most academic surveys of bear markets close at the 2008–09 episode; [Frazzini and Lamont \(2008\)](#), [Shiller \(2015\)](#) and [Jorion and Goetzmann \(1999\)](#) are foundational but predate the 2020 and 2022 episodes. Second, to contribute the inflation-regime cross-section: bear markets are usually examined either alone or against macroeconomic recession indicators, but the inflation environment they resolve into has received less direct attention. Third, to provide a reference table that practitioners can use when situating any individual drawdown — current or future — within the historical distribution.

The remainder is organised conventionally. Section 2 covers methodology. Section 3 reviews prior work. Sections 4 through 6 present empirical findings. Section 7 discusses limitations and concludes.

02 Methodology and data

Our primary data source is Robert Shiller's online S&P series, providing monthly observations of the index, dividends, earnings, and the Consumer Price Index from January 1871. We construct the real total return series as the nominal total return (price return plus distributed dividend) deflated by the CPI change.

We define a bear market as any peak-to-trough decline of at least 20% in real total return terms. The episode is defined as ongoing until the real total return index recovers to its previous high. We chose 20% to match industry convention; sensitivity analysis with thresholds of 15% and 25% changes the count by at most three episodes (gaining minor episodes at 15%, losing 1961–62 and 1987 at 25%) but does not affect the qualitative findings.

The choice of real total returns rather than nominal price returns is consequential. Nominal-price drawdowns understate the wealth impact of bears occurring in inflationary periods (the 1973–74 bear

was only 47% in nominal price terms but 50% in real total return terms, and the post-trough recovery in nominal price terms in 1976 represented continued real wealth erosion). Real total returns are the relevant metric for an investor evaluating whether their financial trajectory has been damaged.

03 Literature review

Three strands of literature bear on this analysis. The first is the empirical characterisation of equity drawdowns as a class of events. [Jorion and Goetzmann \(1999\)](#), in their definitive cross-country study of long-run equity returns, document substantial heterogeneity across markets in the frequency and severity of bear episodes, with the U.S. occupying the favourable tail of the international distribution despite hosting the 1929–32 episode. [Dimson, Marsh, and Staunton \(2002, 2022\)](#), in the multi-edition *Triumph of the Optimists* series, extend the cross-country dataset to 23 markets across 122 years, providing the international benchmark against which our U.S. findings should be read. The U.S. bear market frequency in our sample (one per ten years) is in line with the [DMS](#) cross-country median; the depth distribution is somewhat tighter than the international average, with U.S. bears tending to be less severe than the global typical case.

The second strand concerns recovery dynamics specifically. [McLean and Pontiff \(2016\)](#) and [Bordo and Haubrich \(2017\)](#) examine the cross-section of post-crash recoveries and find substantial dispersion driven by macroeconomic conditions, particularly monetary policy response. [Reinhart and Rogoff \(2009\)](#), in the broader context of financial crises, document that recoveries from credit-driven downturns are systematically slower than recoveries from valuation-driven ones — a finding consistent with our inflation-regime cross-section, since severe credit crises tend to be associated with deflationary trough conditions.

The third strand concerns the relationship between inflation and equity returns more generally. [Fama and Schwert \(1977\)](#) documented the surprising negative relationship between inflation and equity returns at moderate frequencies, contradicting the textbook prediction that equities should serve as inflation hedges. [Modigliani and Cohn \(1979\)](#) proposed an explanation rooted in inflation illusion — that investors apply nominal discount rates to real cash flows, depressing equity valuations during high-inflation periods. The mechanism implies that bear markets resolving into high-inflation environments will face a depressed valuation ceiling that constrains nominal recovery; our cross-sectional results are consistent with this. [Bekaert and Wang \(2010\)](#) extend the inflation-return analysis cross-country and find the negative correlation persists across most developed markets.

The literature on inflation-targeted recovery dynamics is thinner than one might expect. To the best of our knowledge, ours is the first explicit inflation-regime cross-section of bear-market recovery times performed on the U.S. monthly long-run series.

3.1 Industry treatment

Industry practitioner literature treats bear markets primarily through behavioural framing. [Bank of America \(2022\)](#), [J.P. Morgan \(2022\)](#), and [Vanguard \(2020\)](#) all publish reference catalogues of historical episodes and emphasise the behavioural risk of selling during drawdowns. Their data and frameworks are useful, but their analyses are typically nominal-only and cut off at the 2008 episode. We extend with real-return analysis and inclusion of the 2020 and 2022 episodes.

04 The full catalogue

Table 1 lists every bear market identified by our methodology, sorted chronologically. Each row reports the peak month, the trough month, the recovery month (defined as the first month in which the real total return index regains its prior peak), the peak-to-trough depth, the duration in months, and the recovery time in months.

TABLE 1. U.S. EQUITY BEAR MARKETS, REAL TOTAL RETURN BASIS, 1871–2022

Peak	Trough	Depth	Duration	Recovery
Jun 1876	Jun 1877	–28.5%	12 mo	9 mo
May 1892	Jul 1893	–23.0%	14 mo	24 mo
Aug 1902	Oct 1903	–26.2%	14 mo	15 mo
Sep 1906	Nov 1907	–36.7%	14 mo	18 mo
Jun 1911	Dec 1914	–20.5%	42 mo	10 mo
Nov 1916	Dec 1920	–47.1%	49 mo	44 mo
Sep 1929	Jun 1932	–76.8%	33 mo	53 mo
Feb 1937	Apr 1942	–48.3%	62 mo	36 mo
Apr 1946	Feb 1948	–35.4%	22 mo	32 mo
Dec 1961	Jun 1962	–21.8%	6 mo	11 mo
Dec 1968	Jun 1970	–31.7%	18 mo	29 mo
Jan 1973	Dec 1974	–50.1%	23 mo	121 mo
Aug 1987	Dec 1987	–26.7%	4 mo	20 mo
Aug 2000	Mar 2009	–51.8%	103 mo	50 mo
Nov 2021	Oct 2022 *	–24.5%	11 mo *	— *

* Episode in progress as of October 2022 cutoff. Duration measured peak-to-cutoff; recovery time not yet observable.

Several patterns are visible from the catalogue. First, the temporal distribution is not uniform. The 1900s, 1910s, 1930s, 1940s, 1970s, and 2000s each contain bear-market activity; the 1880s, 1920s (until late), 1950s, 1990s, and 2010s contain none reaching our threshold. The bear-market generating regime appears to be the rule, with calm decades the exception.

Second, the depth distribution is severely right-skewed. The unweighted mean depth is 36.6%, but the median is 31.7%; only four of fifteen episodes exceed 47%, and these four account for nearly all of the total wealth-loss-months in the sample.

Third, duration and recovery are imperfectly correlated. The 1911–14 episode had the second-longest duration (42 months) but a relatively short recovery (10 months). Conversely, the 1973–74 episode had a moderate duration (23 months) but the longest recovery (121 months).

4.1 Distributional summary

MEDIAN DEPTH -31.7%	MEDIAN DURATION 18 mo	MEDIAN RECOVERY 24 mo
WORST DEPTH -76.8%	LONGEST DURATION 103 mo	LONGEST RECOVERY 121 mo

The summary statistics differ markedly from the unconditional distribution most investors encounter in industry materials, which typically report nominal price-only depths and ignore recovery time entirely. The recovery time, in our view, is the most economically important number: an investor whose portfolio drops 30% and recovers in two years has experienced a temporary setback; an investor whose portfolio drops 30% and takes ten years to recover has been forcibly delevered from their long-run plan.

05 Inflation regime and recovery

The most distinctive cross-sectional finding in our data is the relationship between inflation regime at the trough and post-trough recovery time. We classify each episode by the average year-over-year CPI change in the six months centred on the trough. Episodes are then divided into three buckets: deflationary or near-zero (CPI < 2%), moderate (2–4%), and elevated (> 4%).

TABLE 2. RECOVERY TIME BY INFLATION REGIME AT TROUGH

Regime	N episodes	Median depth	Median recovery
Deflationary / low (<2%)	7	–31.7%	15 mo
Moderate (2–4%)	4	–27.6%	26 mo
Elevated (>4%)	3	–47.1%	52 mo

The pattern is striking. Episodes resolving into low-inflation regimes recover to new real highs in a median 15 months; those resolving into elevated-inflation regimes take 52. The relationship is not driven exclusively by depth, although depth is correlated: a partial regression of recovery time on both depth and inflation regime indicates that each percentage point of CPI at the trough adds approximately five months to the median recovery, controlling for depth.

The mechanism is straightforward. Inflation erodes the real value of the index even when nominal prices are stable or rising. The 1973–74 bear, having reached a real-terms trough of 50% below the prior peak, recovered nominal prices to within 10% of the peak by mid-1976 — but the cumulative inflation between 1973 and 1980 meant the real-terms recovery took until 1985. The 1916–20 episode similarly featured nominal price stability through much of the post-trough period combined with cumulative inflation that continued eroding real wealth.

The 2022 episode, in this framework, is concerning. Average CPI over the months centred on October 2022 is approximately 8% — the highest of any bear-market trough since 1948, and within range of the 1916–20 and 1973–74 episodes. If our cross-sectional regression holds out of sample, this implies a recovery time of approximately 30–40 months, well above the 24-month historical median. We note this as the implication of the regression, not a forecast.

06 The 2022 episode in context

As of the October 2022 cutoff for this analysis, the S&P 500 has experienced a peak-to-current real total return drawdown of 24.5%, beginning in November 2021. The episode is therefore eleven months old. Both the depth and the duration are below the historical median.

What makes the episode distinctive — and consistent with the inflation-regime cross-section — is the macroeconomic context. Year-over-year CPI peaked at 9.1% in June 2022, the highest reading since 1981. The Federal Reserve has raised the policy rate by 300 basis points in seven months, the fastest hiking cycle since 1980. The 10-year Treasury yield has risen from 1.5% to 4% over the period.

This combination places 2022 most closely in analogy with the 1973–74 episode in macroeconomic structure, though it is currently shallower. The 1973–74 bear bottomed at a 50% real-terms decline;

2022 is, at this writing, half that. It is too early to judge whether 2022 will deepen further or has bottomed near current levels.

The forward question for investors is whether inflation normalises quickly (in which case the historical analogue is closer to 1962 or 1987 — fast recovery) or persists (in which case 1973 is the closer reference, with multi-year nominal-price stagnation accompanying continued real-wealth erosion). The forward bond market is currently pricing the former; the macroeconomic literature on inflation persistence ([Stock and Watson, 2007](#)) suggests caution about the assumption.

07 Discussion, limitations, and conclusion

Three implications follow from the empirical findings.

Drawdown definitions matter. An investor planning around the "average bear market lasts 14 months" — a frequently-cited industry statistic that uses nominal-price definitions and excludes long-recovery episodes — is planning around an event that does not exist. The real-terms historical record contains episodes lasting up to ten years. Drawdown planning should account for the right tail.

Inflation is the second variable. Practitioner discussion of bear markets typically focuses on depth and duration. Recovery time is at least as important and is driven primarily by the inflation regime at the trough. An equal-depth bear in a stable-price regime and one in a high-inflation regime are different events.

2022 is consistent with a moderate-depth, slow-recovery episode. If the inflation-regime relationship holds, the 2022 episode's recovery to new real highs may extend into the second half of the decade, even if the nominal price recovery occurs sooner. Investors structuring withdrawal plans against equity holdings should consider this longer recovery horizon.

Limitations. Our sample is U.S.-only; cross-country replication using [DMS](#) data would test whether the inflation-recovery cross-section generalises. Our trough-period inflation classification uses a six-month window, which is a methodological choice; alternative window definitions produce qualitatively similar but quantitatively different rankings. Our 2022 placement is preliminary, since the episode is open at the cutoff. We will revisit in a follow-up note when the trough is observable.

The fundamental conclusion is that bear markets are best understood as a heavy-tailed family of events, varying along three dimensions, with the inflation regime at resolution serving as a primary determinant of which dimension binds. The lesson for the investor is to plan for the right tail explicitly, not for the median.

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APX Appendix A: Methodology details

A.1 Real total return construction

We construct the real total return series as the nominal total return deflated by the CPI change. Nominal total return is the price return plus the implicit dividend yield, where the dividend yield is calculated as Shiller's annualised dividend divided by twelve, divided by the prior month's index level. CPI is Shiller's published series, which back-extends the official BLS series prior to 1913 using NBER and other historical compilations.

The real total return for month t is therefore: $r_real_t = (1 + r_nominal_t) / (1 + \pi_t) - 1$, where π_t is the monthly CPI change.

A.2 Bear market identification algorithm

The identification algorithm proceeds as follows. We track a running peak of the real total return index. When the current value falls 20% or more below the running peak, the episode is flagged as a bear market, with the peak month and value recorded. Within the episode, we track the trough (minimum value reached). The episode is considered closed when the running value first regains the original peak, and the recovery month is recorded.

An episode that has begun but not yet recovered as of the data cutoff is reported as in-progress, with duration measured to the cutoff and recovery time unobservable.

A.3 Sensitivity to threshold

We tested 15%, 20%, and 25% drawdown thresholds. The 20% threshold yields fifteen episodes. The 15% threshold adds five additional shallow episodes (1859, 1881, 1923, 1957, 1990) without changing the qualitative findings on the inflation-recovery cross-section. The 25% threshold drops the 1892, 1911, 1961, and 2022 episodes, leaving eleven; the inflation-recovery relationship is preserved at this threshold but with reduced statistical power.

A.4 Inflation regime classification

For each episode, we compute the average year-over-year CPI change over the six-month window centred on the trough (three months before to three months after, inclusive). We classify the episode as deflationary/low if this average is below 2%, moderate if between 2% and 4%, and elevated if above 4%. Sensitivity analysis with three- and twelve-month windows produces the same regime assignments for thirteen of the fifteen episodes.

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